

Ramadhan Adam Zome

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EDUCATION

Pan African University (PAUSTI), Nairobi <i>MSc Artificial Intelligence and Machine Learning</i> Supervised by Dr. Albert Njoroge Kahira & Dr. Jude Mwenda	2025 – present
Kenyatta University, Nairobi <i>BSc Statistics and Programming — First Class Honours</i>	2019 – 2023

RESEARCH INTERESTS

Self-supervised representation learning · AI security · open-world malware detection · low-resource NLP · philosophy of mind

EXPERIENCE

Data Analyst & Cybersecurity Associate — Innovative Toll Solutions, Nairobi — Data cleaning, EDA, and modelling on operational logs and transaction datasets — Built dashboards and reporting pipelines used by operations teams — Log-level anomaly detection, alert triage, and traffic pattern identification — Supported pipeline resilience work following DDoS incidents	Dec 2023 – Jul 2025
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SELECTED PROJECTS

Open-World Windows Malware Detection MSc thesis research. Self-supervised byte-level embeddings with prototypical few-shot adaptation for detecting novel malware families in open-world conditions.	2025 – present
Deepfake Detection + RAG Explainability Xception CNN trained from scratch on 140k real/fake faces. RAG pipeline (FAISS + sentence-transformers + LLM) for natural-language model explanations. FastAPI backend on HuggingFace Spaces, React/Vite frontend.	2025 – 2026
BirdSpeak — Acoustic Token Transformer Transformer trained on MFCC + k-means quantized acoustic tokens for bird call classification. Framed vocalization understanding as an NLP sequence modelling problem. Trained on Xeno-canto data.	2025
Swahili GPT Character-level GPT trained from scratch on 65M characters of Swahili news text (29,544 articles). Explores low-resource language modelling for an underrepresented African language.	2025
Neural Networks from First Principles (C) Implementations of foundational models in C: Elman RNN with genetic algorithm training and a standalone raylib visualizer. Emphasis on understanding internals over framework use.	2024 – present
Malware Analysis Lab Writeups Practical static and dynamic analysis of real malware samples including Poison Ivy. PE analysis, anti-VM evasion techniques, and sandboxing. Published as a technical series on Medium.	2026

TECHNICAL SKILLS

Languages: Python, C, SQL, Bash
ML / DL: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, FAISS

Security: Malware analysis, reverse engineering, PE static analysis, sandboxing (REMnux, FLARE-VM)

Tools: Git, Linux, Docker, FastAPI, React/Vite, Jupyter

RELEVANT COURSEWORK

Deep Learning · Natural Language Processing · Computer Vision & Pattern Recognition · Robotics · Ethics in AI · Probability & Statistics · Algorithms

References available upon request.